

CII SR -14th Edition Kaizen Compétition 2019

A Kaizen Presentation By



EVEREST INDUSTRIES LIMITED
(Podanur Works – Coimbatore)

Company Profile

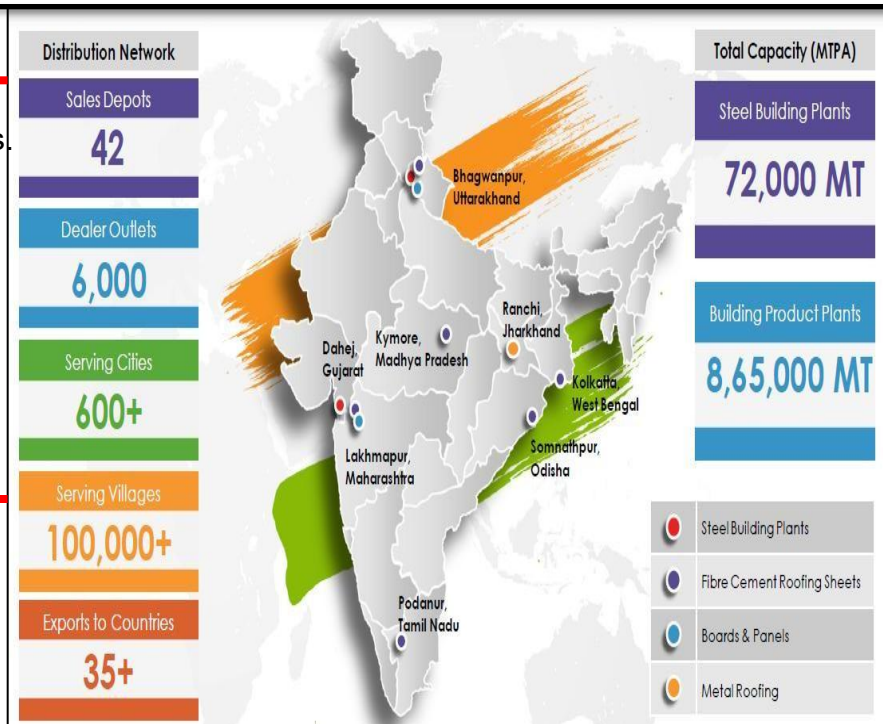


Company Overview:

Everest Industries Limited, incorporated in 1934, has a rich history in manufacturing of Building products and Steel products. Everest offers a complete range of roofing ceiling, wall, flooring and cladding products distributed through a large network, and also pre-engineered steel buildings for industrial, commercial and residential applications. It is one of the leading building solution providers in India, providing detailed technical assistance in the form of design, drawings and implementation for every project.

Our Business:

- **Building Products (63%)** – include fiber cement roofing sheets, fiber cement boards, solid wall panels.
- **Steel buildings (37%)** – offers customized building solutions like Pre-Engineered Steel buildings, metal roofing sheets and cladding.



Fibre Cement Roofing Sheets



Hi-Tech Roofing Sheets



Super Sheets



Cement Planks



Fibre Cement Boards



Rapicon Walls



Smart Steel Buildings



Our Esteemed Customers



Kaizen Theme

Kaizen Theme: To reduce Consumption of Tele-Break (Anti-Foaming) Oil.

Implemented Area: Sheeting Machine-1 Shop Floor, Podanur Works, Coimbatore

Implementation Date: September 2018

Cross Functional Team:



Team Leader
Mr. Y. Kesavaraj
(Line-In charge, Production)



Team Leader
Mr. K. Mani
(Line-In charge, Production)



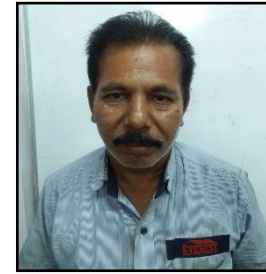
Mr. M. Vijay Kumar
(Senior Engineer Elect. Maint)



Mr. N. Karthikeyan
(Engineer - Production)



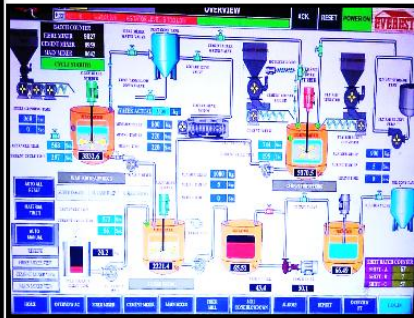
Mr. N. Anil Kumar
(Engineer - QS & BE)



Mr. Edward
(Fitter)

Problem / Present Status

Process Flow:



**RAW MATERIAL
PREPARATION**



CONE TANK



**VATS & SIEVE
FILTRATION**



**BACK WATER
TRENCH**



SHEET FORMING



DE-PIILING



**SHEET
HANDLING**



PRE CURING

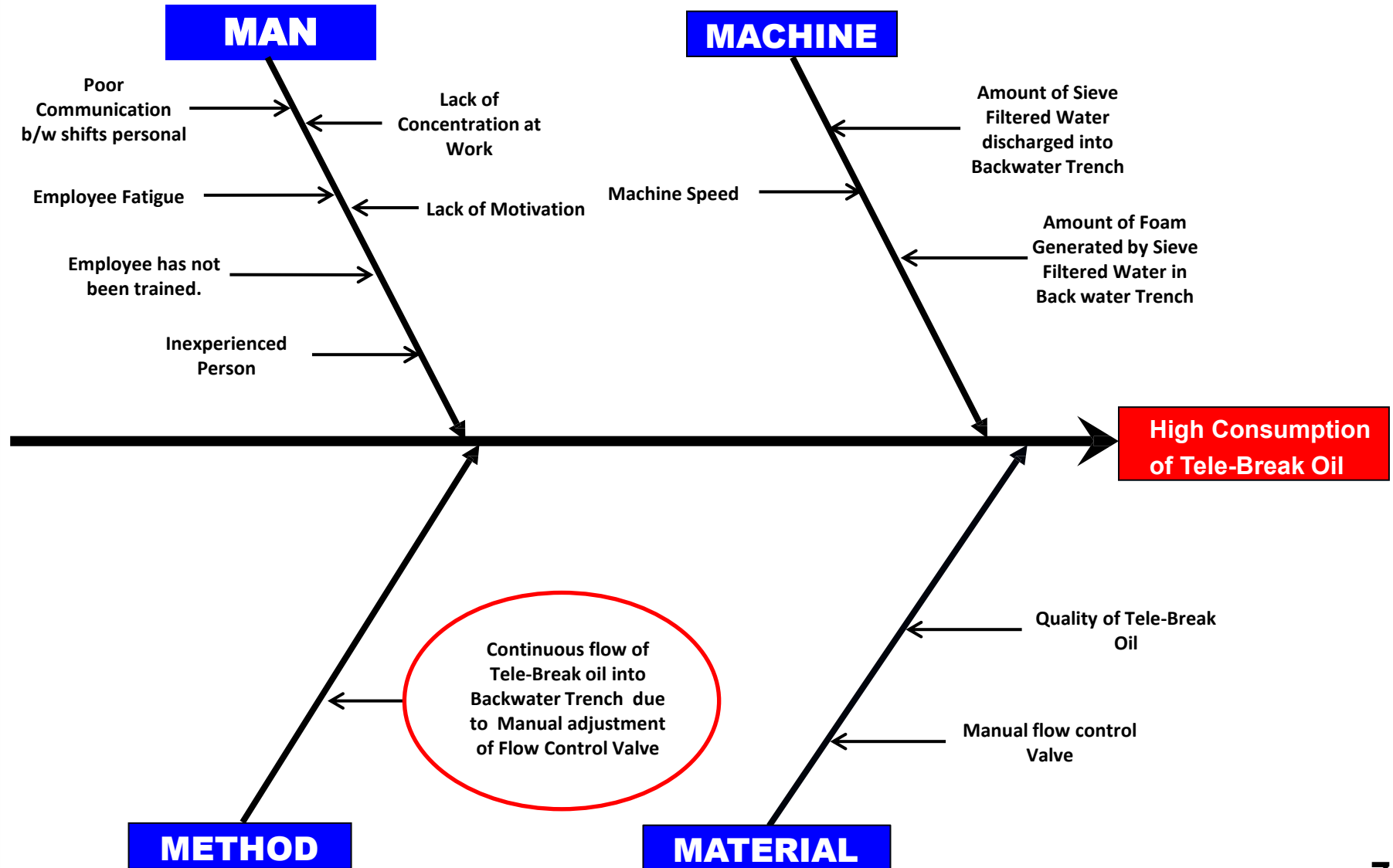
Problem / Present Status



Problem/ Present Status :

More Consumption of Tele-Break (Anti-Foaming) Oil in our process due to continuous flow of Tele-Break Oil into backwater trench because the flow of Tele-Break Oil is controlled by adjusting manual flow valve.

Cause & Effect Analysis



Root Cause Analysis & Idea Generation



Problem: More consumption of Tele-Break (Anti-Foaming) Oil in our process.

Why 1 : Continuous flow of Tele-Break oil into process.

Why 2 : Flow of Tele-Break oil is controlled manually.

Why 3 : There is NO Automatic Flow control system.

Root Cause: No Automatic control system to control the flow of Tele-Break oil into process.

Idea Generation: To provide PLC Control for Tele-Break oil flow system.

Action: Provided PLC Control to Tele-Break oil flow system with timing arrangement.

Countermeasures

SN	Counter Measure Activities	Target Date	Responsibility	Status
01	To study Tele-Break(Anti-Foaming) oil quantity required for controlling the Foam generation based on time.	01/09/2019	Production Department	Completed
02	Provide PLC Control to Tele-Beak oil flow system based on timing arrangement.	05/09/2019	Maintenance Department	Completed



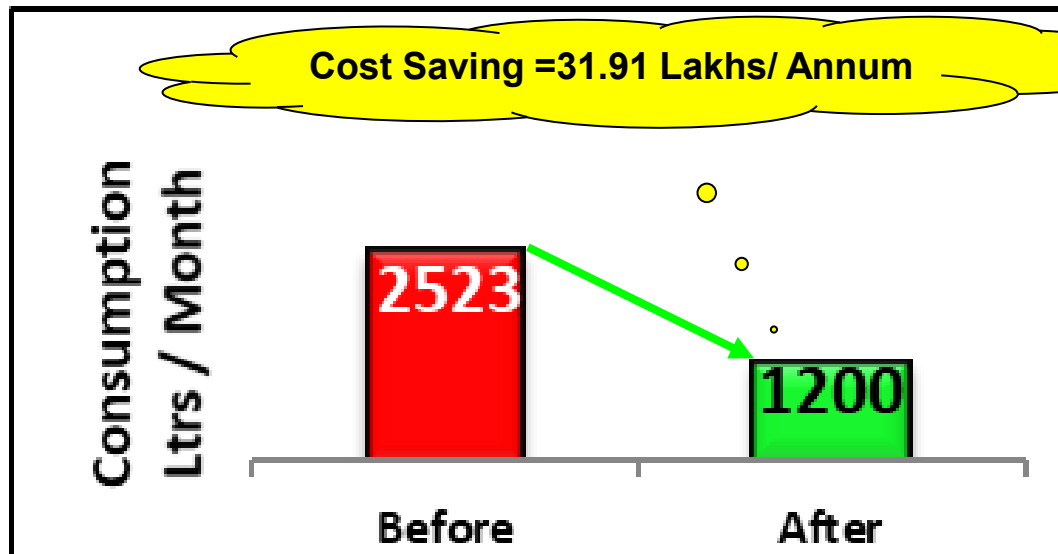
- Single Solenoid Valve with PLC automation system implemented to control the Tele-Break oil consumption, based on Delay timer arrangement in PLC.



- Single Solenoid Valve with PLC automation system implemented for Tele-Break oil to control the flow, based on Delay timer arrangement in PLC.

Validation of Data: Calculation

Before		After	
Average Monthly Consumption of Anti-Foaming Oil	2523 Ltrs	Average Monthly Consumption of Anti-Foaming Oil	1200 Ltrs
Cost of Tele-Break Oil	201 Rs/Ltr	Cost of Tele-Break Oil	201 Rs/Ltr
Monthly Cost Consumption of Tele-Break Oil	5,07,123 Rs/Month	Monthly Cost Consumption of Tele-Break Oil	2,41,200 Rs/Month
Yearly Cost Consumption of Tele-Break Oil	60,85,476 Rs/Year	Yearly Cost Consumption of Tele-Break Oil	28,94,400 Rs/Year



Tangible Benefits

- Average Tele-Break Oil Saving of 15876 Ltrs/ Year.
- Cost saving of 31,91,076 Rs/ Year.
- Flow of the Tele-Break oil in process is controlled automatically without any Human intervention.

In-tangible Benefits

- Proper usage of Resources.
- Consumption of Natural Resources has reduced.
- Morale of People increased due to reduction in consumption of Tele-Break oil and Cost saving by simple improvement.

- Training on Standard Operating Procedures are given to all the maintenance people for regular inspection & maintenance.

Tele-Break (Anti-Foaming) Oil Standard Operating Procedure

1. On the starting of each shift check the level of Tele-Break Oil in the Tank. If the level of Tele-Break oil is low fill the Tank up to the required level.
2. Check the flow of Tele-Break oil into Backwater for any abnormal conditions.
3. Check delay timing arrangement on the starting of each shift to know if there is change in machine settings.
4. If any changes have made by previous Shift-In charges reset the settings.
5. Check the level of Tele-Break (Anti-Foaming) oil in the Tank, flow of oil into Backwater trench once for every hour.

Kaizen Sustenance:

WHAT TO DO	Regular Inspection and Maintenance.
HOW TO DO	Regular Training and Standardized Check sheet.
FREQUENCY	Weekly.

Horizontal Deployment:

BW	CW	KW	LW	PW1	PW2	SW

Kaizen Sheet

Kaizen Sheet							EVEREST		Company	MM/YY	S.No												
Productivity	Quality	Delivery	Safety	Material	Energy	Maintenance	Everest Industries Ltd 09/18 2																
Kaizen Title: To reduce consumption of Tele-Break(Anti-Foaming) Oil.							Implemented Area: PW Shopfloor																
Problem/Present Status: More consumption of Tele-Break (Anti-foaming) oil in our process.				Before Improvement: 			Implemented by: Y. Kesavaraj, K. Mani, M. Gnanaprakash, M. Vijay Kumar, N. Anil Kumar, N. Karthikeyan, Edward																
Root Cause Identification: Problem •More consumption of Tele-Break oil. Why1 •Continuous flow of Tele-Break oil into process. Why2 • Flow of Tele-Break oil is controlled manually. Why3 •There is NO Automatic Flow control system.				After Improvement: 			Result/Benefit: (a) Qualitative Proper usage of resources.																
							(b) Quantitative 																
Root cause				No automatic control system to control the flow of Tele-Break oil into process.			Standardization: Preventive Maintenance as per schedule.																
Idea to eliminate root cause				To provide PLC control for Tele-Break oil flow system.			Horizontal Deployment:																
Counter-measure				Single Solenoid Valve with PLC automation system implemented to Tele-Break oil for control the flow, based on Delay timer arrangement in PLC.			<table border="1" style="width: 100%; text-align: center;"> <tr> <td>BW</td> <td>CW</td> <td>KW</td> <td>LW</td> <td>PW</td> <td>SW</td> </tr> <tr style="background-color: green;"> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> </table>					BW	CW	KW	LW	PW	SW						
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Thank You